# Seokin Seo (서석인)

Curriculum Vitae

Postdoctoral Researcher (from Mar. 2026)
Advanced Robot Control Lab (advisor: *Yisoo Lee*)
Center for Humanoid Research,
KIST (Korea Institute of Science and Technology)

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## Education

**KAIST**, Ph.D. in Kim Jaechul Graduate School of AI, 2026.
▶ **Dissertation:** Robust Imitation Learning under Data Collection Biases (co-advised by *Kee-Eung Kim*, *Hongseok Yang*)

**KAIST**, M.S. in School of Computing, 2019. (advised by *Kee-Eung Kim*)
▶ **Thesis:** A Study on Generating Explanations for Reinforcement Learning Policies in Control Tasks

**Sogang University**, B.S. in Computer Science & Mathematics, 2017.

## Research Interests

**Reinforcement Learning, Imitation Learning, Physical AI**
Reliable decision-making from imperfect data, deployed in real-world systems.

## Patents

**Regularized Behavior Cloning for Blocking the Leakage of Past Action Information**
KR Patent Application, No. 10-2023-0159062, Nov. 2023.

## Awards

**Gold Reviewer Award**
International Conference on Machine Learning (ICML), Jul. 2026.

**Outstanding Paper Award**
Korea Software Congress (KSC), Feb. 2023.

## Academic Experiences

**Guest Lecturer**

Mathematics for AI (AI503), KAIST (Fall 2025): *Distribution Shift in IL*

**Teaching Assistant**

Introduction to RL, KAIST Machine Learning Engineer Bootcamp (2022)
Mathematics for AI (AI503), KAIST (Fall 2021)
Dialogue System, KAIST–Samsung DS AI Expert Program (2020)
Machine Learning (CS376), KAIST (Fall 2019)
Data Structure (CS206), KAIST (Spring/Fall 2018)

**Academic Talk**

Robust and Safe Offline Imitation Learning via Distribution Matching, Locomotion and Manipulation Technology Workshop, Korea University (2026)

Mitigating Covariate Shift in Behavioral Cloning via Robust Stationary Distribution Correction, AI Tutorial Series, KAIST (2025)

**Academic Reviewer**

NeurIPS (2021, 2024, 2026), ICLR (2025, 2026), ICML (2025, 2026), ICRA (2027)

## Programming Skills

**Languages & Frameworks**

Python (PyTorch, JAX), C, C++, Ruby (Rails), Java, JavaScript
Anaconda, Docker, Git, Kubernetes, Nomad

**Simulator Experiences**
MuJoCo (MJX, MJWarp), IsaacGym, IsaacSim, CARLA

## Publication List (*: equal contribution, †: equal advising)

### International Conference

[I9] **Seokin Seo**, Sung Kuk Shyn, Minji Seo, Yisoo Lee†, Jongmin Lee†, Hongseok Yang†, Kee-Eung Kim†.
*DexDICE: Reconciling Safety and Performance via Dual-Expert Offline Imitation Learning.*
40th Conference on Neural Information Processing Systems (NeurIPS), 2026.

[I8] Yoonseok Choi, Chaeyoung Oh, Hyunjun Choi, **Seokin Seo**, Kee-Eung Kim.
*Concept Removal Guidance: Evidence-Calibrated Negative Guidance for Safe Diffusion Sampling.*
Proceedings of the 43rd International Conference on Machine Learning (ICML), 2026. **(Spotlight)**

[I7] Joo Bon Maeng*, Seongmin Lee*, **Seokin Seo**, Kee-Eung Kim.
*Goal-Conditioned DPO: Prioritizing Safety in Misaligned Instructions.*
Annual Conference of the Nations of the Americas Chapter of the ACL (NAACL), 2025.

[I6] **Seokin Seo**, Byung-Jun Lee, Jongmin Lee, HyeongJoo Hwang, Hongseok Yang, Kee-Eung Kim.
*Mitigating Covariate Shift in Behavioral Cloning via Robust Stationary Distribution Correction.*
38th Conference on Neural Information Processing Systems (NeurIPS), 2024.

[I5] **Seokin Seo**, HyeongJoo Hwang, Hongseok Yang, Kee-Eung Kim.
*Regularized Behavior Cloning for Blocking the Leakage of Past Action Information.*
37th Conference on Neural Information Processing Systems (NeurIPS), 2023. **(Spotlight)**

[I4] HyeongJoo Hwang, **Seokin Seo**, Youngsoo Jang, Sungyoon Kim, Geon-Hyeong Kim, Seunghoon Hong, Kee-Eung Kim.
*Information-Theoretic State Space Model for Multi-View Reinforcement Learning.*
Proceedings of the 40th International Conference on Machine Learning (ICML), 2023. **(Oral)**

[I3] Geon-Hyeong Kim, **Seokin Seo**, Jongmin Lee, Wonseok Jeon, HyeongJoo Hwang, Hongseok Yang, Kee-Eung Kim.
*DemoDICE: Offline Imitation Learning with Supplementary Imperfect Demonstrations.*
International Conference on Learning Representations (ICLR), 2022.

[I2] Youngsoo Jang, **Seokin Seo**, Jongmin Lee, Kee-Eung Kim.
*Monte-Carlo Planning and Learning with Language Action Value Estimates.*
International Conference on Learning Representations (ICLR), 2021.

[I1] Wonseok Jeon, **Seokin Seo**, Kee-Eung Kim.
*A Bayesian Approach to Generative Adversarial Imitation Learning.*
32nd Conference on Neural Information Processing Systems (NeurIPS), 2018. **(Spotlight)**

### Preprints & Under Review

[P2] Sol Choi, Mingon Kim, **Seokin Seo**, Jongeun Choi, Donghyeon Kim, Yisoo Lee.
*Reusing Locomotion Training Recipes Across Humanoids via Body-Aligned Virtual Joints.*
Under review, 2026.

[P1] Seongmin Lee*, Jaewook Shin*, Young Jin Ahn, **Seokin Seo**, Ohjoon Kwon, Kee-Eung Kim.
*Zero-Shot Multi-Hop Question Answering via Monte-Carlo Tree Search with Large Language Models.*
arXiv preprint arXiv:2409.19382, 2024.

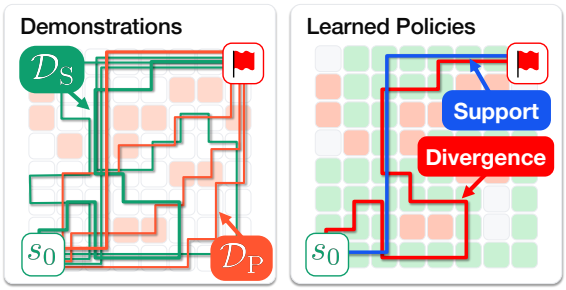
### Domestic Conference & Journals

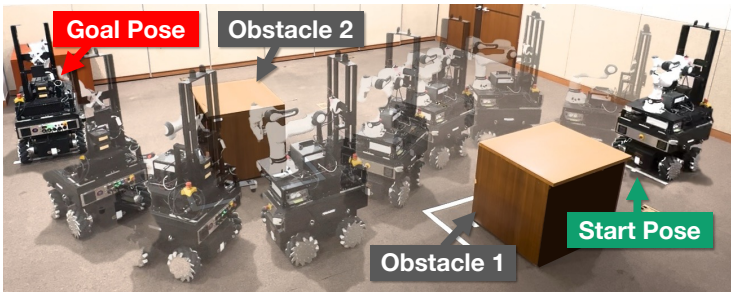
[D2] **Seokin Seo**, HyeongJoo Hwang, Hongseok Yang, Kee-Eung Kim.
*Improving Causally-Regularized Logistic Regression via Confounder Balancing with Feature Combinations.*
Korea Software Congress (KSC), 2022. **(Outstanding Paper Award)**

[D1] Geon-Hyeong Kim, **Seokin Seo**, Shinhyung Kim, Kee-Eung Kim.
*Variational Recurrent Neural Networks with Relational Memory Core Architectures.*
Journal of KIISE, Vol. 47, No. 2, pp. 189–194, 2020.

## Selected Publications

**1. DexDICE**: Safe Offline IL with Dual-Expert Demonstrations — NeurIPS 2026 · [I9]



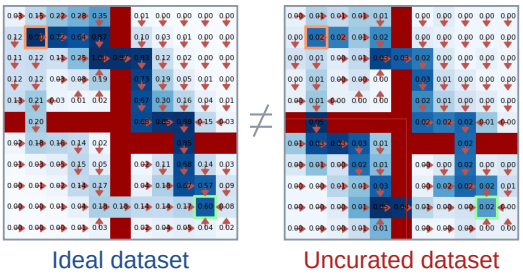




- Safe IL typically requires a cost function that is hard to specify; we instead learn **safe behavior without a cost function** from two expert sets, performance-relevant $\mathcal{D}_P$ and safety-relevant $\mathcal{D}_S$.
- We optimize a DICE-based offline IL objective that keeps the agent close to $\mathcal{D}_P$ while staying within the support of $\mathcal{D}_S$.
- Evaluated on safe RL benchmarks (DSRL) and a real-world mobile navigation task under a range of constraints.

**Benefit.** Safety requirements are often available only as examples of acceptable behavior; this makes them learnable from logged data.

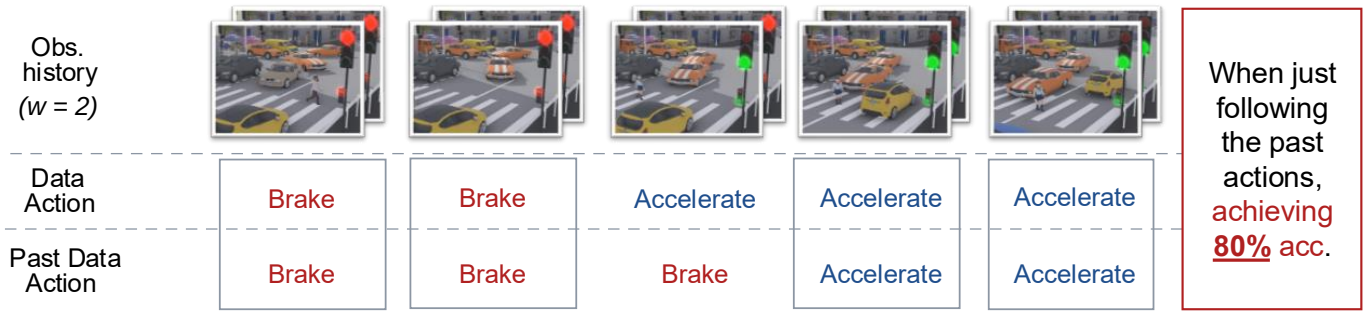
**2. DrilDICE**: Robust Offline IL under Covariate-Shifted Expert Datasets — NeurIPS 2024 · [I6]

- Prior IL methods assume the dataset follows the expert's stationary distribution, which realistic data collection (segmented trajectories, collection bias) often violates even with expert-quality actions.
- We make offline IL robust to **covariate shift** by formulating it as distributionally robust optimization and deriving a practical algorithm, DrilDICE, via an offline RL interpretation.
- Evaluated on continuous-control benchmarks with covariate-shifted datasets, where DrilDICE consistently outperforms BC and offline IL baselines.




**Benefit.** Operational logs often cover only fragments of a system's full operating horizon; DrilDICE improves robustness to the resulting data-collection bias.

**3. PALR**: Mitigating Action Leakage in History-Based IL — NeurIPS 2023 **Spotlight** · [I5]




- We study IL from observation histories, commonly used under partial observability (e.g., vision-based control) to capture velocity and acceleration.
- Offline IL in this setting induces **action leakage**, a strong dependence on past actions; we quantify it and propose a regularized BC approach to mitigate it.
- Evaluated on state-based MuJoCo tasks and vision-based driving tasks in CARLA simulator.

**Benefit.** Deployed controllers act on sensor histories under partial observability; this identifies a failure mode specific to that setting and fixes it.